Bond Prices, Yields, and Arbitrage

- We're going to study the yield curve
- Specifically, we're going to study
  1. What the yield curve can tell us about the economy
  2. What current yields tell us about future yields
  3. How to price bonds using the yield curve

Arbitrage

- Consider a 1-year, 100 USD, zero-coupon bond that compounds annually
- Suppose that you can borrow and lend for 1 year at 10% (which compounds annually)
- Now

$$p_0 \overset{\text{“should”}}{=} \frac{100 \text{ USD}}{1 + 0.1} = 90.9091 \tag{1}$$

- Suppose that $p_0 \overset{\text{“actual”}}{=} 100$.

Arbitrage

- Suppose that $p_0 \overset{\text{“actual”}}{=} 90$.

Security	$\mathbf{CF}_0$	$\mathbf{CF}_1$
Bond	-90.0000	$+100$
Borrow	$+90.9091$	-100
	$+0.9091$	0

- On date 0 (today), borrow 90.9091 USD, use 90 of it to buy the bond, pocket the .9091. On date 1 (in one year), collect the 100 from the bond and use it to pay off your loan.

Arbitrage

- Suppose that $p_0 \overset{\text{"actual"}}{=} 91$.

Security	$\mathbf{CF}_0$	$\mathbf{CF}_1$
Bond	$+91.0000$	-100
Lend	-90.9091	$+100$
	0.9091	0

- On date 0 (today), short the bond, collect 91 USD, lend 90.9091 of it, and pocket the .9091. On date 1 (in one year), collect the 100 USD from your loan and pay 100 to the person whose bond you shorted.

- These portfolios were "free" in the sense that they did not require any up-front capital
- If one trader observes the mispricing, she will be able to exploit it and earn arbitrage profits
- As more and more traders enter the market to buy (or sell) the bond, the price converges to its "should" price

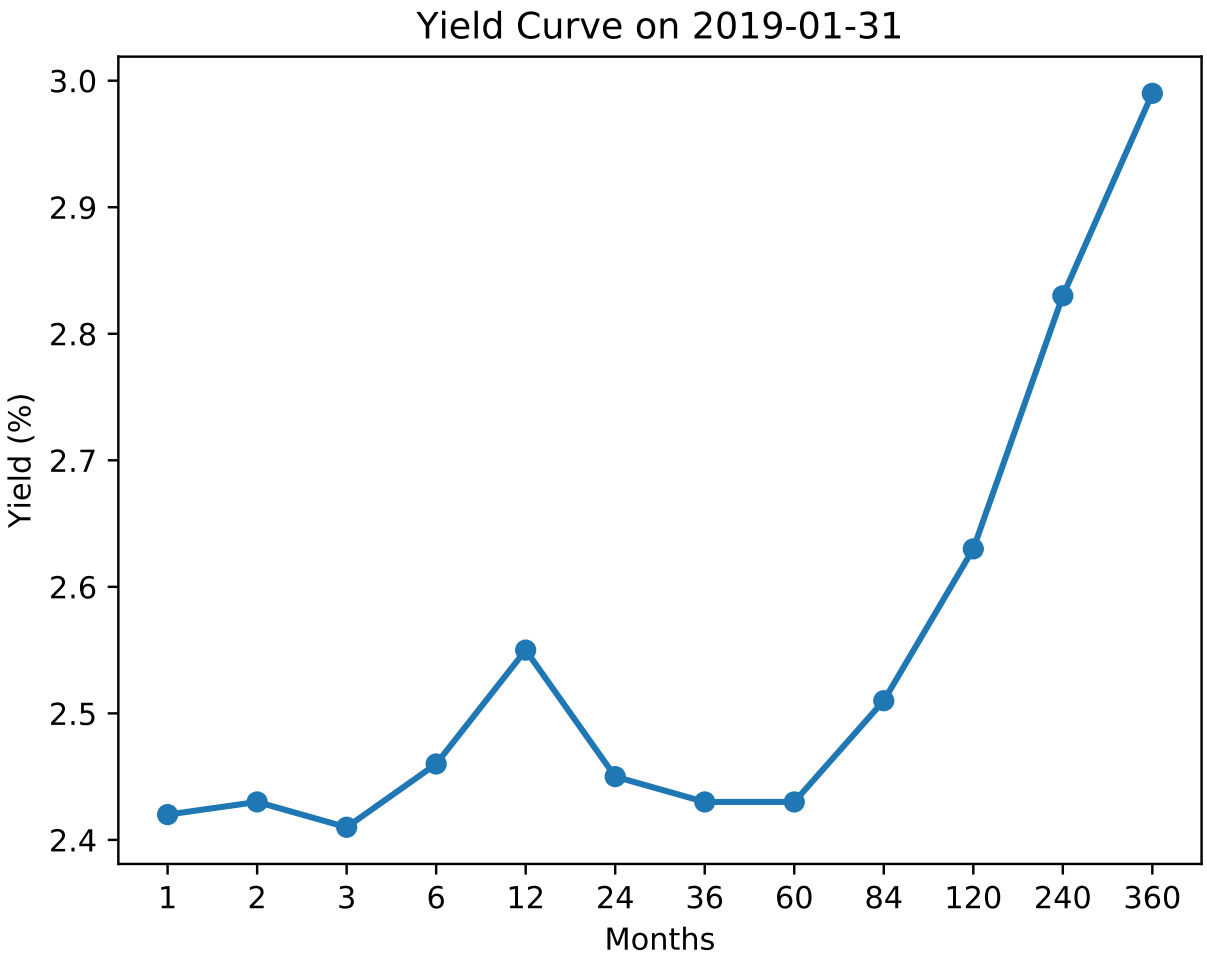
- For each term (e.g. 1 month, 1 year, 30 years), the yield curve tells us the yield of the corresponding zero-coupon bond
- Example 1: if the 10-year yield is 1.61% (as read from the yield curve), then a 10-year zero coupon bond should have yield 1.61% and should trade for

$$P = \frac{100 \text{ USD}}{1 + 10 \times 0.0161} = 86.1326 \text{ USD} \tag{2}$$

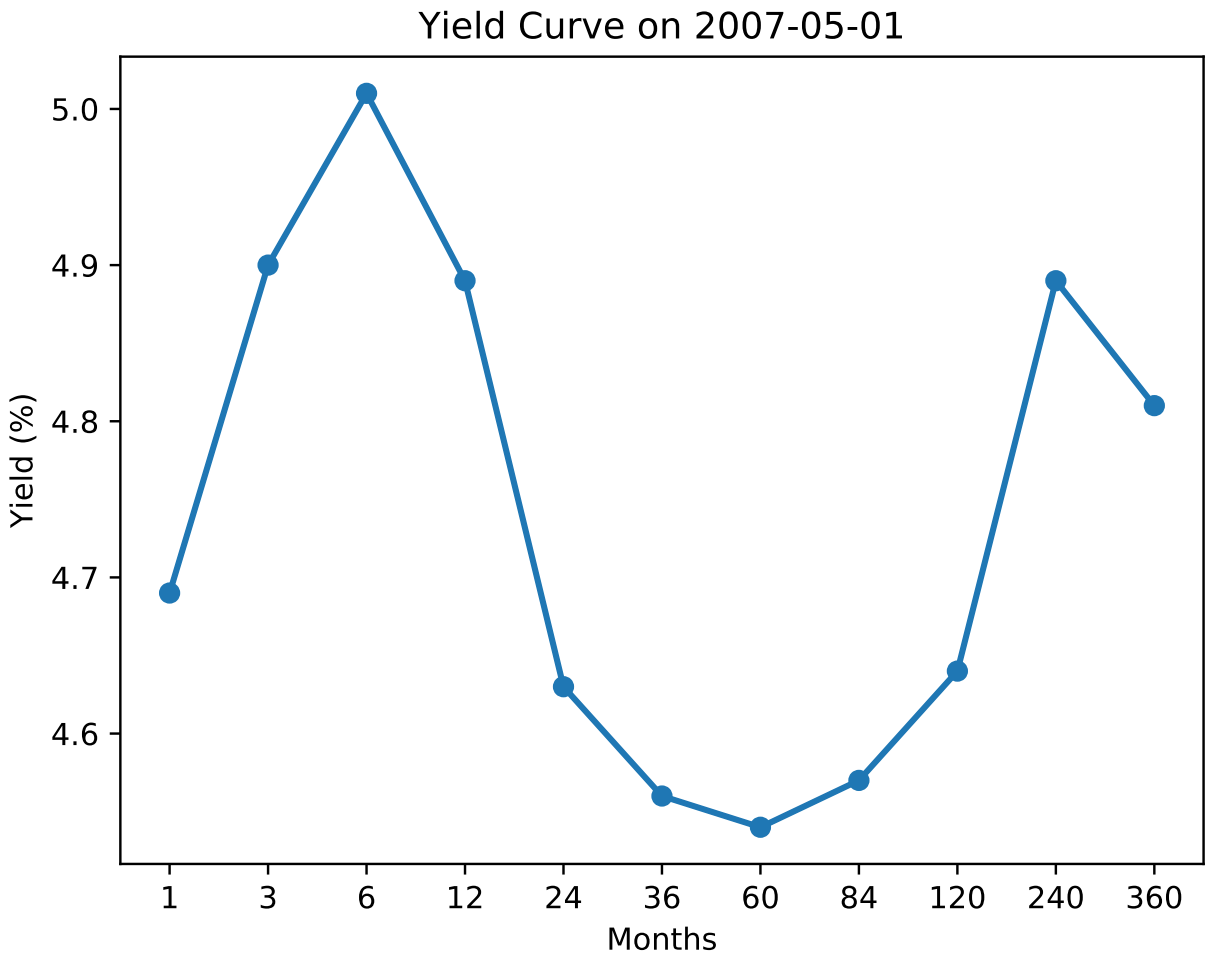
- It is <u>not</u> the yield on a 10-year Treasury note (because the note's yield depends on its coupon rate)
- Example 2: if the 1-month yield is 1.55% (as read from the yield curve), then a 1-month zero coupon bond should have yield 1.55% and should trade for

$$P = \frac{100 \text{ USD}}{1 + 0.0155/12} = 99.8710 \text{ USD} \tag{3}$$

A typical yield curve



Entering recession

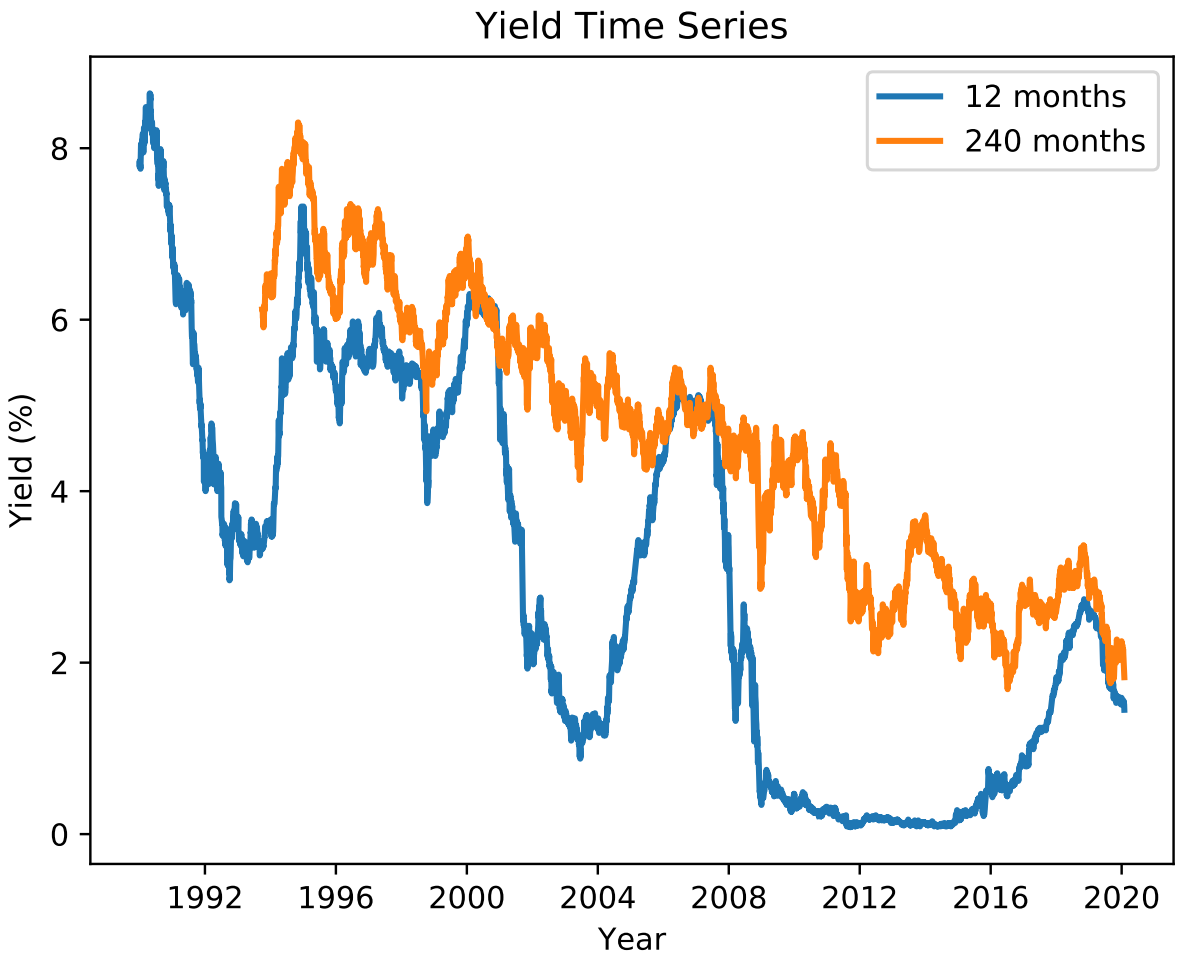


Why are there different rates?

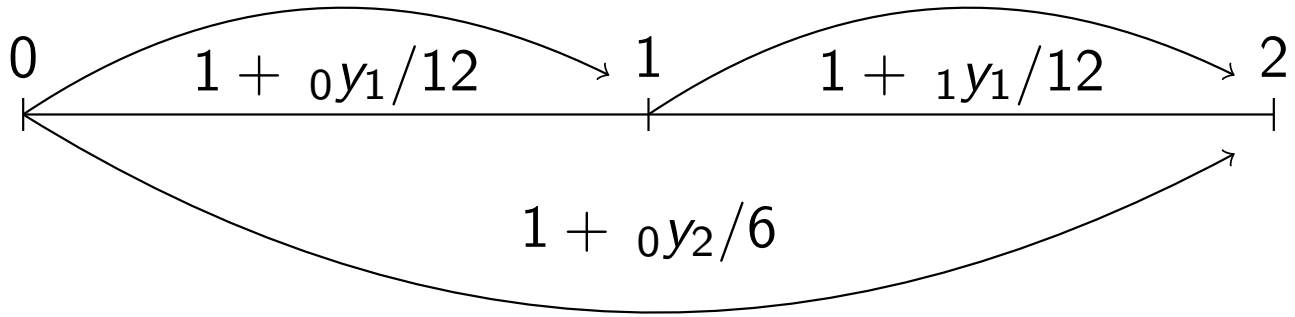
- Why are there different rates? Why isn't there just one rate for all terms?
- The shape of the yield curve is pinned-down by something called a *stochatic discount factor*, the discussion of which is beyond the scope of this course[1]
- When conditions are expected to improve, the curve is upward sloping; when they are expected to deteriorate, the curve is downward sloping
- Application: banks borrow short and lend long

[1] The interested student should study *Optimal Control Theory*

In the time-series...

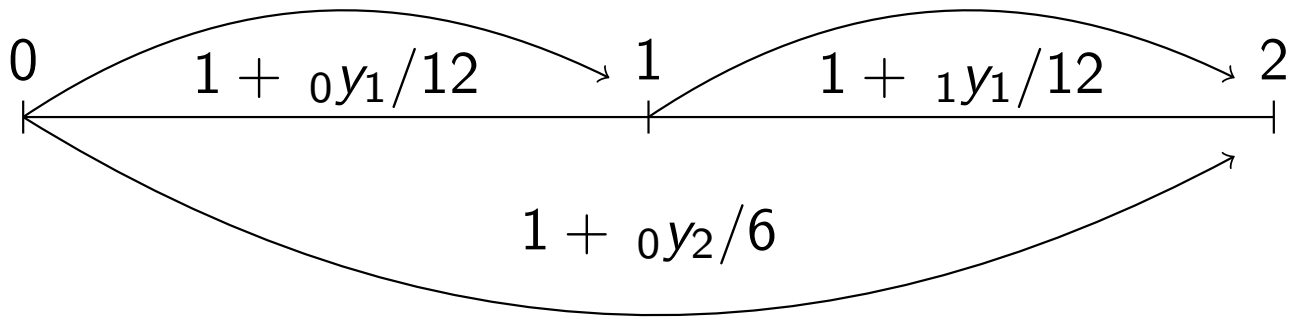


Forward Rates (One-Month)



- ${}_0y_1$ is the yield today (date 0) on a 1-month zero coupon bond that matures in one month (date 1)
- ${}_0y_2$ is the yield today (date 0) on a 2-month zero coupon bond that matures in two months (date 2)
- ${}_1y_1$ is the yield in one month (date 1) on a 1-month zero coupon bond that matures in two months (date 2)

Forward Rates (One-Month)



$$(1 + {}_0y_1/12)(1 + {}_1y_1/12) \overset{\text{should}}{=} 1 + {}_0y_2/6. \tag{4}$$

- The current yield curve $({}_0y_1, {}_0y_2)$ predicts the future short-term yield $({}_1y_1)$, which we call the *forward rate*
- This relationship between current yields and future yields is called the *expectations hypothesis*

Forward Rates

${}_0y_1 = 1.57\%$ and ${}_0y_2 = 1.58\%$. Is the market optimistic or pessimistic about the economy (in the short-term)? What did the market expect the 1-month yield to be in one month (${}_1y_1$)?

Forward Rates

${}_0y_1 = 1.57\%$ and ${}_0y_2 = 1.58\%$. Is the market optimistic or pessimistic about the economy (in the short-term)? What did the market expect the 1-month yield to be in one month (${}_1y_1$)?

$$ {}_1y_1 = 12\left(\frac{1 +_0 y_2/6}{1 +_0 y_1/12} - 1\right) = 1.5879\% \tag{5} $$

Forward Rates

On 2019-01-01, ${}_0y_1 = 2.40\%$ and ${}_0y_2 = 2.40\%$. What did the market expect the 1-month yield to be in one month (${}_1y_1$)?

Forward Rates

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$$ {}_1y_1 = 12\left(\frac{1 +_0 y_2/6}{1 +_0 y_1/12} - 1\right) = 2.3952\% \tag{6} $$

Coupon Bonds

We can use the yield curve to compute the price of a coupon bond. A T-year coupon bond makes annual payments

0	1	2	$\cdots$	T
$-P_{\text{cpn}}$	cF_{cpn}	cF_{cpn}	$\cdots$	$cF_{\text{cpn}} + F_{\text{cpn}}$

where $c \geq 0$ is the coupon rate (APR) and F_{cpn} is the face value of the coupon bond

$$P_{\text{cpn}} \overset{\text{should}}{=} \left[\frac{cF_{\text{cpn}}}{F_{1\text{-yr}}}\right] P_{1\text{-yr}} + \left[\frac{cF_{\text{cpn}}}{F_{1\text{-yr}}}\right] P_{2\text{-yr}} + \cdots + \left[\frac{(1+c)F_{\text{cpn}}}{F_{T\text{-yr}}}\right] P_{T\text{-yr}}$$

where y is the yield (APR) and $P_{k\text{-yr}}$ is the price of a k-year zero coupon bond with face value of $F_{k\text{-yr}}$

Example

What is the price of a 3-year, 1000 USD 10% coupon bond that makes payments annual coupon payments. The 1-year yield is 1.48%, the 2-year yield is 1.41%, and the 3-year yield is 1.40%. What should the 3-year *coupon* bond trade for?

Example

What is the price of a 3-year, 1000 USD 10% coupon bond that makes payments annual coupon payments. The 1-year yield is 1.48%, the 2-year yield is 1.41%, and the 3-year yield is 1.40%. What should the 3-year *coupon* bond trade for? First, compute the prices of the zero-coupon bonds:

$$p_1 = \frac{100 \text{ USD}}{1 + 1 \times 0.0148} = 98.5416 \text{ USD} \tag{7}$$

$$p_2 = \frac{100 \text{ USD}}{1 + 2 \times 0.0141} = 97.2373 \text{ USD} \tag{8}$$

$$p_3 = \frac{100 \text{ USD}}{1 + 3 \times 0.0140} = 95.9693 \text{ USD} \tag{9}$$

Next, note that the cash-flows from the zero coupon bond are 100 USD, 100 USD, and 1100 USD.

$$p_{\text{cpn}} = 1 \times p_1 + 1 \times p_2 + 11 \times p_3 = 1251.4612 \text{ USD} \tag{10}$$

Example

What is the price of a 3-year, 1000 USD 5% coupon bond that makes payments annual coupon payments. The 1-year yield is 1.59%, the 2-year yield is 1.58%, and the 3-year yield is 1.62%. What should the 3-year *coupon* bond trade for?

Example

What is the price of a 3-year, 1000 USD 5% coupon bond that makes payments annual coupon payments. The 1-year yield is 1.59%, the 2-year yield is 1.58%, and the 3-year yield is 1.62%. What should the 3-year *coupon* bond trade for? First, compute the prices of the zero-coupon bonds:

$$p_1 = \frac{100 \text{ USD}}{1 + 1 \times 0.0159} = 98.4349 \text{ USD} \tag{11}$$

$$p_2 = \frac{100 \text{ USD}}{1 + 2 \times 0.0158} = 96.9368 \text{ USD} \tag{12}$$

$$p_3 = \frac{100 \text{ USD}}{1 + 3 \times 0.0162} = 95.3652 \text{ USD} \tag{13}$$

Next, note that the cash-flows from the zero coupon bond are 50 USD, 50 USD, and 1050 USD.

$$p_{\text{cpn}} = \frac{1}{2} \times p_1 + \frac{1}{2} \times p_2 + \left(10 + \frac{1}{2}\right) \times p_3 = 1099.0205 \text{ USD} \tag{14}$$

Yield-to-Maturity

- In our previous lectures, we saw that market participants borrow/lend at different yields *depending on the term*
- Rather than carry around an entire yield curve whenever we want to recompute the price of a bond, we summarize this information with the *yield-to-maturity* (YTM), with which you are familiar from your previous coursework
- Example: consider a 1000 USD, 30-year, 5% T-bond that makes semi-annual payments, and trades for 1011.9610 USD
- The YTM on this T-bond is the <u>one</u> rate that would make the PV of cash-flows equal to the price, namely, 4.9233%

YTM Example

Suppose that 1, 2, and 3-year, 100 USD zero-coupon bonds trade for 106.4607, 99.7684, and 93.7438 USD respectively. What should be the YTM of a 3-year, 1000 USD, 10% coupon bond that makes annual payments?

YTM Example

Suppose that 1, 2, and 3-year, 100 USD zero-coupon bonds trade for 106.4607, 99.7684, and 93.7438 USD respectively. What should be the YTM of a 3-year, 1000 USD, 10% coupon bond that makes annual payments?

First compute what the price should be.

$$p_{\text{cpn}} \overset{\text{should}}{=} 1\times 106.4607 + 1\times 99.7684 + 11\times 93.7438 = 1237.4109 \text{ USD}$$

Then we can compute the YTM using a financial calculator: NPER = 3, PMT = 100 USD, PV = 1237.4109 USD, and FV = 1000 gives us a YTM of 1.7998%.